

TOUCH

The sense of touch comes from microscopic sensory receptors (specialized endings of nerve cells) in the skin or in deeper tissues (see p.182). Some receptors are enclosed in capsules of connective tissue, while others are uncovered. Different shapes and sizes

of receptors detect a range of stimuli, such as light touch, heat, cold, pressure, and pain. The receptors relay their signals via the spinal cord and lower brain to a strip curving around the cerebral cortex, known as the somatosensory cortex, or “touch centre”.

Epidermis

Layer of constantly renewing cells; multiply at base; harden and die as they move outward

Superficial nerve endings

Occur everywhere in the skin

Meissner's corpuscle

Upper dermal nerve ending; mostly located just below epidermis

Merkel's disc receptor

Junction nerve ending; sited just above or below the boundary between epidermis and dermis

SENSORY RECEPTORS

When a sensory receptor in the skin is deformed, or temperature changes make it expand or contract, it generates nerve impulses.

Ruffini corpuscle

Nerve ending, mostly in middle or lower dermal layers

Dermis

Skin layer housing most touch receptors

Pacinian corpuscle

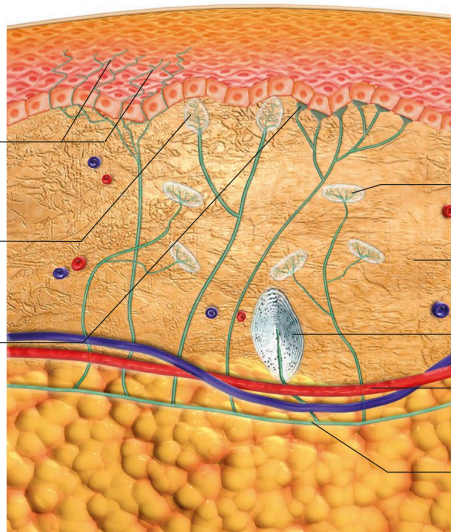
Located deep in the dermis

Blood vessel

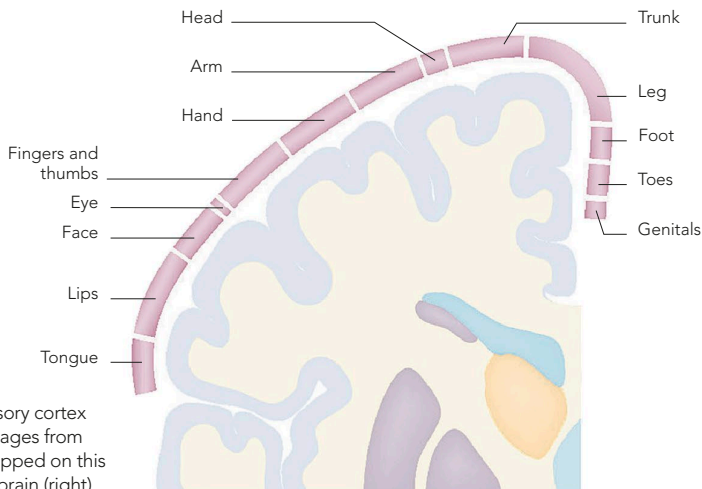
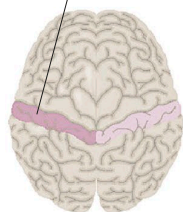
Brings nourishment to the skin layers and touch receptors

Nerve fibre

Receptors' nerve fibres gather into bundles; these convey signals to the main nerves



Left somatosensory cortex



TOUCH MAP

Each part of the somatosensory cortex (above) receives touch messages from skin around the body, as mapped on this vertical cross-section of the brain (right).